

Instructions: There are totally 2 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (6 points) Use Laplace transform to solve the initial value problem:

$$x'' + 16x = 0; \quad x(0) = 4, \quad x'(0) = 5.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\mathcal{L}\{x'(t)\} = sX(s) - 4$$

$$\mathcal{L}\{x''(t)\} = s^2X(s) - 4s - 5$$

Thus, $\mathcal{L}\{x'' + 16x\} = \mathcal{L}\{0\}$ gives us

$$\mathcal{L}\{x''\} + 16\mathcal{L}\{x\} = 0$$

$$[s^2X(s) - 4s - 5] + 16X(s) = 0$$

$$(s^2 + 16)X(s) = 4s + 5$$

$$X(s) = \frac{4s + 5}{s^2 + 16}$$

$$\begin{aligned} \Rightarrow x(t) &= \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{4s + 5}{s^2 + 16}\right\} = \mathcal{L}^{-1}\left\{\frac{4s}{s^2 + 16}\right\} + \mathcal{L}^{-1}\left\{\frac{5}{s^2 + 16}\right\} \\ &= 4\cos 4t + \frac{5}{4}\sin 4t \end{aligned}$$

Problem 2. (4 points) Find the inverse transform of $\frac{1}{s(s-4)}$ using the formula $\mathcal{L}^{-1}\left(\frac{F(s)}{s}\right) = \int_0^t \mathcal{L}^{-1}(F(s))d\tau$.

$$\begin{aligned} \text{Solution: } \mathcal{L}^{-1}\left\{\frac{1}{s(s-4)}\right\} &= \int_0^t \mathcal{L}^{-1}\left\{\frac{1}{s-4}\right\}d\tau \\ &= \int_0^t e^{4\tau}d\tau \\ &= \left[\frac{1}{4}e^{4\tau}\right]_{\tau=0}^t \\ &= \frac{1}{4}(e^{4t} - 1) \end{aligned}$$